

EXPERIMENT-3

AIM:

Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.

QUERY:

1. Create Database and Tables

```
CREATE DATABASE CompanyDB;
USE CompanyDB;

-- Department Table
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50) NOT NULL,
    Location VARCHAR(50)
);

-- Project Table
CREATE TABLE Project (
    Project_ID INT PRIMARY KEY,
    Project_Name VARCHAR(100) NOT NULL,
    Budget DECIMAL(12,2),
    Dept_ID INT,
    FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID)
);

-- Employee Table
CREATE TABLE Employee (
    Emp_ID INT PRIMARY KEY,
    Emp_Name VARCHAR(100) NOT NULL,
    Salary DECIMAL(10,2),
    Job_Title VARCHAR(50),
    Dept_ID INT,
    Project_ID INT,
    FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID),
    FOREIGN KEY (Project_ID) REFERENCES Project(Project_ID)
);
```

2. Insert 5 Departments

```
INSERT INTO Department VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Mumbai'),
(3, 'Finance', 'Bangalore'),
(4, 'Marketing', 'Pune'),
(5, 'Operations', 'Chennai');
```

3. Insert 8 Projects

```
INSERT INTO Project VALUES
(101, 'Website Development', 500000, 1),
(102, 'Mobile Application', 750000, 1),
(103, 'Recruitment System', 300000, 2),
```

```
(104, 'Financial Analysis', 600000, 3),
(105, 'Marketing Campaign', 450000, 4),
(106, 'Customer Management', 550000, 4),
(107, 'Inventory System', 400000, 5),
(108, 'Business Automation', 800000, 5);
```

4. Insert 30 Employees

INSERT INTO Employee VALUES

```
(1, 'Aarav Sharma', 55000, 'Developer', 1, 101),
(2, 'Riya Gupta', 62000, 'Developer', 1, 102),
(3, 'Arjun Verma', 48000, 'Tester', 1, 101),
(4, 'Ananya Singh', 70000, 'Team Lead', 1, 102),
(5, 'Karan Mehta', 45000, 'Support Engineer', 1, 101),
(6, 'Priya Sharma', 58000, 'Developer', 1, 102),
(7, 'Neha Kapoor', 50000, 'HR Executive', 2, 103),
(8, 'Rahul Jain', 65000, 'HR Manager', 2, 103),
(9, 'Simran Kaur', 42000, 'Recruiter', 2, 103),
(10, 'Aditya Roy', 46000, 'HR Executive', 2, 103),

(11, 'Megha Joshi', 55000, 'Recruiter', 2, 103),
(12, 'Vivek Malhotra', 72000, 'HR Manager', 2, 103),
(13, 'Ishita Rao', 60000, 'Accountant', 3, 104),
(14, 'Manish Kumar', 75000, 'Financial Analyst', 3, 104),
(15, 'Pooja Agarwal', 52000, 'Accountant', 3, 104),
(16, 'Rohit Bansal', 68000, 'Financial Analyst', 3, 104),
(17, 'Kavya Nair', 58000, 'Accountant', 3, 104),
(18, 'Sahil Gupta', 80000, 'Finance Manager', 3, 104),
(19, 'Tanya Singh', 47000, 'Marketing Executive', 4, 105),
(20, 'Yash Verma', 56000, 'Marketing Analyst', 4, 105),

(21, 'Nisha Patel', 62000, 'Marketing Manager', 4, 106),
(22, 'Aman Khan', 44000, 'Sales Executive', 4, 106),
(23, 'Shreya Das', 51000, 'Marketing Executive', 4, 105),
(24, 'Dev Sharma', 59000, 'Sales Executive', 4, 106),
(25, 'Akash Yadav', 49000, 'Operations Executive', 5, 107),
(26, 'Sneha Roy', 57000, 'Operations Analyst', 5, 107),
(27, 'Varun Singh', 63000, 'Operations Manager', 5, 108),
(28, 'Komal Gupta', 46000, 'Operations Executive', 5, 107),
(29, 'Nitin Sharma', 54000, 'Process Analyst', 5, 108),
(30, 'Aditi Mehta', 69000, 'Operations Manager', 5, 108);
```

SQL Queries

5. Selection

Selection retrieves rows that satisfy a condition.

Employees having salary greater than ₹60,000

SELECT *

FROM Employee

WHERE Salary > 60000;

Emp_ID	Emp_Name	Salary	Job_Title	Dept_ID	Project_ID
2	Riya Gupta	62000.00	Developer	1	102
4	Ananya Singh	70000.00	Team Lead	1	102
8	Rahul Jain	65000.00	HR Manager	2	103
12	Vivek Malhotra	72000.00	HR Manager	2	103
14	Manish Kumar	75000.00	Financial Analyst	3	104
16	Rohit Bansal	68000.00	Financial Analyst	3	104
18	Sahil Gupta	80000.00	Finance Manager	3	104
21	Nisha Patel	62000.00	Marketing Manager	4	106
27	Varun Singh	63000.00	Operations Manager	5	108
30	Aditi Mehta	69000.00	Operations Manager	5	108

6. Projection

Projection retrieves only specific columns from a table.

SELECT Emp_ID, Emp_Name, Salary
FROM Employee;

Emp_ID	Emp_Name	Salary
1	Aarav Sharma	55000.00
2	Riya Gupta	62000.00
3	Arjun Verma	48000.00
4	Ananya Singh	70000.00
5	Karan Mehta	45000.00
6	Priya Sharma	58000.00
7	Neha Kapoor	50000.00
8	Rahul Jain	65000.00
9	Simran Kaur	42000.00
10	Aditya Roy	46000.00
11	Megha Joshi	55000.00
12	Vivek Malhotra	72000.00
13	Ishita Rao	60000.00
14	Manish Kumar	75000.00
15	Pooja Agarwal	52000.00
16	Rohit Bansal	68000.00
17	Kavya Nair	58000.00
18	Sahil Gupta	80000.00
19	Tanya Singh	47000.00
20	Yash Verma	56000.00
21	Nisha Patel	62000.00
22	Aman Khan	44000.00
23	Shreya Das	51000.00
24	Dev Sharma	59000.00
25	Akash Yadav	49000.00
26	Sneha Roy	57000.00
27	Varun Singh	63000.00
28	Komal Gupta	46000.00
29	Nitin Sharma	54000.00
30	Aditi Mehta	69000.00

7. Aggregate Functions

COUNT, SUM, AVG, MAX, MIN

SELECT

COUNT(*) AS Total_Employees,

SUM(Salary) AS Total_Salary,

AVG(Salary) AS Average_Salary,

MAX(Salary) AS Highest_Salary,

MIN(Salary) AS Lowest_Salary

FROM Employee;

Total_Employees	Total_Salary	Average_Salary	Highest_Salary	Lowest_Salary
30	1718000.00	57266.666667	80000.00	42000.00

8. GROUP BY

Find the average salary of each department:

SELECT Dept_ID, AVG(Salary) AS Average_Salary

FROM Employee

GROUP BY Dept_ID;

Dept_ID	Average_Salary
1	56333.333333
2	55000.000000
3	65500.000000
4	53166.666667
5	56333.333333

Using the department name:

SELECT d.Dept_Name, COUNT(e.Emp_ID) AS Employee_Count

FROM Department d

JOIN Employee e

ON d.Dept_ID = e.Dept_ID

GROUP BY d.Dept_Name;

Dept_Name	Employee_Count
IT	6
HR	6
Finance	6
Marketing	6
Operations	6

9. HAVING

HAVING is used to filter groups created by GROUP BY.

Departments having more than 5 employees

SELECT Dept_ID, COUNT(*) AS Employee_Count

FROM Employee

GROUP BY Dept_ID

HAVING COUNT(*) > 5;

Dept_ID	Employee_Count
1	6
2	6
3	6
4	6
5	6

10. CASE Expression

Classify employees according to their salary:

SELECT

Emp_Name, Salary,

CASE

WHEN Salary >= 70000 THEN 'High Salary'

WHEN Salary >= 50000 THEN 'Medium Salary'

ELSE 'Low Salary'

END AS Salary_Category

FROM Employee;

Emp_Name	Salary	Salary_Category
Aarav Sharma	55000.00	Medium Salary
Riya Gupta	62000.00	Medium Salary
Arjun Verma	48000.00	Low Salary
Ananya Singh	70000.00	High Salary
Karan Mehta	45000.00	Low Salary
Priya Sharma	58000.00	Medium Salary
Neha Kapoor	50000.00	Medium Salary
Rahul Jain	65000.00	Medium Salary
Simran Kaur	42000.00	Low Salary
Aditya Roy	46000.00	Low Salary
Megha Joshi	55000.00	Medium Salary
Vivek Malhotra	72000.00	High Salary
Ishita Rao	60000.00	Medium Salary
Manish Kumar	75000.00	High Salary
Pooja Agarwal	52000.00	Medium Salary
Rohit Bansal	68000.00	Medium Salary
Kavya Nair	58000.00	Medium Salary
Sahil Gupta	80000.00	High Salary
Tanya Singh	47000.00	Low Salary
Yash Verma	56000.00	Medium Salary
Nisha Patel	62000.00	Medium Salary
Aman Khan	44000.00	Low Salary
Shreya Das	51000.00	Medium Salary
Dev Sharma	59000.00	Medium Salary
Akash Yadav	49000.00	Low Salary
Sneha Roy	57000.00	Medium Salary
Varun Singh	63000.00	Medium Salary
Komal Gupta	46000.00	Low Salary
Nitin Sharma	54000.00	Medium Salary
Aditi Mehta	69000.00	Medium Salary

11. ORDER BY**Sort employees by salary in ascending order****SELECT Emp_Name, Salary****FROM Employee****ORDER BY Salary ASC;**

Emp_Name	Salary
Simran Kaur	42000.00
Aman Khan	44000.00
Karan Mehta	45000.00
Aditya Roy	46000.00
Komal Gupta	46000.00
Tanya Singh	47000.00
Arjun Verma	48000.00
Akash Yadav	49000.00
Neha Kapoor	50000.00
Shreya Das	51000.00
Pooja Agarwal	52000.00
Nitin Sharma	54000.00
Megha Joshi	55000.00
Aarav Sharma	55000.00
Yash Verma	56000.00
Sneha Roy	57000.00
Kavya Nair	58000.00
Priya Sharma	58000.00
Dev Sharma	59000.00
Ishita Rao	60000.00
Nisha Patel	62000.00
Riya Gupta	62000.00
Varun Singh	63000.00
Rahul Jain	65000.00
Rohit Bansal	68000.00
Aditi Mehta	69000.00
Ananya Singh	70000.00
Vivek Malhotra	72000.00
Manish Kumar	75000.00
Sahil Gupta	80000.00